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Hybrid explainable machine learning for population level cardiovascular disease risk prediction using satellite environmental data

Ismail Hossain[2], Kazi Md. Tanvir Anzum[1*] and Jarin Tasnim Mumu[1]

*Correspondence:
Kazi Md. Tanvir Anzum
tanvir@iem.kuet.ac.bd
[1]Department of Industrial Engineering and Management, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh
[2]Department of Mechanical Engineering, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

Abstract

Cardiovascular disease (CVD) remains one of the main causes of death worldwide, and better risk prediction is needed to support early prevention and public health planning. Most existing prediction studies focus mainly on clinical variables, while environmental and regional influences are often underused. This study proposes a Hybrid Explainable Artificial Intelligence and Machine Learning (HXAI-ML) framework for area-level CVD risk prediction using health-related and satellite-derived environmental data at the Core-Based Statistical Area (CBSA) level. The dataset comprised 329 observations and 80 variables. Data preprocessing included median imputation, standardization, correlation-based feature filtering, and binary class labeling. To address class imbalance, several resampling strategies were evaluated, including Random Oversampling, SMOTE, Tomek Links, Instance Hardness Threshold, and hybrid combinations of these methods. Five ensemble classifiers were then assessed: Decision Tree, Random Forest, Extra Trees, Gradient Boosting, and Extreme Gradient Boosting. Model performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, Kappa, Matthews Correlation Coefficient, MAE, MSE, and RMSE. Among all tested configurations, SMOTE combined with Extreme Gradient Boosting achieved the strongest performance, with 95.45% accuracy, 97.67% precision, 95.45% recall, and 96.55% F1-score. To enhance transparency, SHAP, LIME, and Permutation Importance Analysis were applied to identify and explain the most influential predictors. Overall, the proposed framework offers a robust and interpretable approach for population-level cardiovascular risk assessment and environmental health decision-making.

Keywords Cardiovascular disease, Risk prediction, Explainable artificial intelligence, Machine learning, Environmental health

1 Introduction

Cardiovascular diseases (CVDs) remain the foremost cause of death globally. The WHO estimates CVDs to kill approximately 18 million people annually. It is almost a third of all deaths in the world [50].). Predominant causes of CVD vary with geographical locations in the world. Such disparities happen because of access to health care, social-economic

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status, and environmental exposures. All these put individuals at a risk of cardiovascular diseases. They develop more cardiovascular illness than there are among those in cleaner environments [22, 45]. The increasing CVD burden is affected by a number of interacting factors. These are the accelerated urbanization, sedentary lifestyles, and environmental exposures [49]. The studies indicate that the environmental and socioeconomic disparities are imminent factors in determining cardiovascular risks. Such differences also have impacts on disease outcomes in dissimilar populations [3]. Cardiovascular diseases can be successfully prevented by early risk recognition. CVD incidence can be minimized with timely treatment and changing their lifestyle towards a healthier one [36]. Environmental factors like long term air pollution exposure can cause cardiovascular abnormalities [24]. In addition to air pollution, several environmental factors like climate conditions, exposure to chemical substance can elevate CVD risk [37]. Management and prevention of environmental risk factors is hence of great essence. Indicatively, a higher quality of air policies resulted in less CVD deaths in various nations. Environmental policies have also been reported to have some positive impact on the cardiovascular health of people [53]. The traditional risk factors such as age, blood pressure, cholesterol and smoking are all clinical risk factors that are typically estimated by predictive models like SCORE or QRISK. But these models do not account for the effects of the environment, or demographic factors, on cardiovascular risk [30]. In recent times, machine learning model have advanced to new opportunities for improving CVD prediction. Hybrid models can improve prediction measurement by leveraging the algorithm's strengths. They can handle data imbalance, provide interpretable insights into the influence of CVD risk [15]. In the context, integrating environmental factors with conventional risk factors and data-driven approach provides support to effective preventive strategies.

1.1 Technical challenges

There are a number of technical challenges involved in the process of integrating a CVD prediction model with the use of heterogeneous healthcare datasets. These problems should be resolved to come up with proper and interpretable results.

1. Data Imbalance and Integration: Adopting data related to environment, satellite data and health-related provides great challenge considering differences in dataset structure, scale and quality. Besides, class imbalance can mislead machine learning models towards the majority class, making prediction of minority data biased.
2. Incorporating Environmental Risk Factors: Environmental, socioeconomic, epidemical factors' access to healthcare services play a significant role in CVDs. However, integrating these factors remains challenging due to spatial granularity and data availability.
3. Limited Interpretability of ML Models: Many advanced machine learning algorithms function as black-box models, making it difficult for clinicians and policymakers to understand how predictions are generated. Lack of interpretability limits the adoption of such models in healthcare decision-making.
4. Generalization for Different Geographic Regions: Even when built from a different population, models' performance across different regions is likely to be limited since environmental exposures, healthcare access, and population characteristics may differ. Achieving adequate performance in a variety of populations is difficult.

1.2 Problem statement

Cardiovascular disease (CVD) is among the most important causes of death in the world and there is a need for reliable estimates of risk. On the other hand, traditional prediction models are based solely on clinical variables and neglect environmental and socioeconomic factors that could contribute to regional cardiovascular risk profiles. Despite advances in machine learning tools for predicting accuracy, most of the research to date has been limited to single clinical parameters and offers little benefit in providing model interpretability. Moreover, satellite-derived environmental variables and spatial health determinants are underutilized to a great degree in the prediction frameworks for CVDs still today. Hence, an integrated and explainable approach combining environmental and health-related data for more comprehensive and transparent cardiovascular risk assessment is needed.

1.3 Research questions

1. Which hybrid data balancing strategy gives the overall best performance on imbalanced rubrics of CVD data?
2. How interpretable and meaningful are the predictions made by the proposed CVD risk prediction model?
3. Can hybrid ML models outclass conventional statistical machine learning models in predicting CVD risk?

To address these questions, this study has some objectives. They are described in the next sub-section.

1.4 Objectives

The study aims to address the challenges faced by the objectives following below:

1. Integrate SAT Health CBSA and Ohio ICD datasets to combine environmental factors with the clinical ones for CVD prediction.
2. Develop an innovative hybrid machine learning algorithm that can predict CVD risk accurately and manage class imbalance and high dimensional data.
3. Evaluate the model to identify key factors using different data balancing techniques influencing cardiovascular disease risk.
4. Integrate XAI techniques to ensure the proposed model provides clinically interpretable predictions.

The novelty of this study does not involve describing an entirely novel machine learning prediction algorithm, but rather the goal of building an integrated explainable prediction framework for the assessment of cardiovascular risk at the area-level. The proposed HXAI-ML framework integrates environmental variables from the CBSA-level, satellite-derived variables with health-related indicators, unlike previous studies using mostly health-related and clinical datasets. Additionally, it systematically investigates several class balancing techniques in response to the well-recognized challenge of imbalanced healthcare datasets. It uses ensemble learning models and integrates SHAP, LIME, and Permutation Importance Analysis to generate both global and local model explanations. This fusion helps provide better predictability with transparency and interpretability in public health applications.

The remainder of the article is organized as follows: Sect. 2 reviews the research literature regarding the prediction of cardiovascular disease risk. Section 3 provides an overview of the suggested methodology and explains it. Section 4 provides an evaluation study of the suggested framework. Section 5 evaluates our method, explains the study topics, and discusses the key findings, and recommendations. Section 6 concludes the paper with a summary of the paper's contributions and limitations.

2 Literature review

The environmental conditions are very critical in determining the health outcomes in humans especially cardiovascular diseases (CVDs). Recent literature indicates that air pollution, climatic variability, availability of green space, and built environment are some of the determinants of cardiovascular risks and prevalence of cardiovascular-related diseases.

2.1 Environmental factors in affecting CVD risk

Green cover and other environmental characteristics are closely associated with the physical health indicators and disease patterns. Climatic conditions and urban design are also important factors associated with population health status. Previous studies indicate that green areas in the city reduce cardiovascular diseases, including ischemic heart disease and stroke. Green spaces promote better lifestyles and reduced exposure to environmental stressors in urban populations [6]. Longitudinal studies also indicate positive correlations between tree cover and positive cardiovascular health outcomes. Urban vegetation has been linked to a lower heart disease, high blood pressure and diabetes [7]. Barboza et al. carried out health impact researches on various cities in Europe. Their findings indicated increased access to green spaces was associated with lower mortality rates. This decrease also comprises those deaths associated with cardiovascular diseases [9]. These findings further beneficial association between natural environments and cardiovascular health. Another prominent environmental risk factor of CVD is air pollution. Air pollution increases blood pressure and the risk of cardiovascular events in the affected people. Strong impact of cardiovascular health of the population is associated by pollutants like particulate matter (PM2.5) and nitrogen dioxide [32]. The environmental pollution may be influenced by the soil and climatic conditions and can affect the concentration of pollutants. Such environmental interactions may contribute to the outcome of human health greatly [12]. The risk of cardiovascular diseases is also determined by factors related to climate variation like variation in temperature. Environmental exposure pathways and physiological stress can be enhanced by the atmospheric conditions. Expansive social and environmental health determinants have been the subject of more studies over the last few years. There have been shown great influence of the neighborhood factors, access to healthcare facilities, and socioeconomic status on the outcomes of the disease. All these are very determinants of the population health and risk of disease [17, 21]. The study by Wafula and colleagues reported that disease prevalence in societies is dependent on environmental and socioeconomic factors [52]. In East Africa, a study was conducted to examine environmental, as well as social, determinants of health. It demonstrated significant associations between community environmental variables and the patterns of infectious and chronic diseases [21]. Other studies of cities find that there are high correlations between the environment and health consequences.

These associations are seen among various communities and groups of people [29]. Advances in satellite data and remote sensing technologies enable large-scale environmental health studies. Researchers have increasingly used satellite images to explore the connection between green spaces and health. These articles also study mental health and disease indicators based on earth observation data [5]. The latest studies also use deep learning models on satellite images. These models can be used to examine the relationships between the built environment and the occurrence of coronary artery disease [13, 14]. Unlike those of cardiometabolic diseases, neighborhood effects on environmental measurements are also studied using satellite-based measurements. Such methods present fresh knowledge about spatial environmental factors contributing to disease development [13, 14]. The urban design has also been pointed out as an important issue in cardiovascular health, as highlighted by Lam et al. The structure of the cities and their neighborhoods has been associated to put a powerful impact on the metabolic syndrome and cardiovascular diseases [26]. Moreover, other environmental factors of the living conditions have been associated to shape the health statuses of individuals. Mobility, physical activity levels, and general wellbeing are factors that influence health status [20].

2.2 ML models in CVD risk prediction

Machine learning (ML) has attracted considerable interest in predicting and diagnosing cardiovascular diseases (CVD). These methods examine complex medical datasets to identify patterns in the case of heart diseases. A wide range of machine learning algorithms has been used in prediction of heart diseases are numerous. Commonly used methods include Support Vector Machine, Random Forest, Naive Bayes, Decision Tree and Logistic Regression. Numerous studies have employed these algorithms to enhance the accuracy of predicting heart diseases [4]. Bashir et al. proposed a majority-vote ensemble model that involves a number of classifiers. The ensemble combined Naive Bayes, Decision Tree, Memory-Based Learner and Support Vector Machine. They tested their method with a number of datasets of the UCI repository. Their model achieved a prediction accuracy of approximately 88.52% [10]. Tomar and Agarwal applied the Least Square Twin Support Vector Machine (LSTSVM) algorithm to examine heart disease by using Statlog data. Their model achieved an accuracy of 85.8% by using the feature selection of F-score to detect important attributes [48]. Decision tree-based approaches have also been widely employed for cardiovascular disease analysis. Fast Decision Tree and pruned C4.5 algorithms were applied by El-Bialy et al. Their models were evaluated using the Cleveland, Hungarian and Statlog heart disease data. The classifier demonstrated promising results with ten-fold cross-validation on the models [18]. Another study by Amin et al. examined the impact of feature selection on a prediction performance. They contrasted various classifiers with the various combinations of the selected features. These classifiers were: KNN, SVM, Logistic Regression and Neural Networks. The analysis identified nine relevant attributes that were important in the performance of prediction. The features chosen generated a mean prediction of 87.41 percent [4]. Ensemble learning techniques have also shown strong results in risk prediction of heart diseases. Mienye et al. developed an accuracy-weighted ensemble model incorporating KNN, Logistic Regression, SVM, Random Forest, and Gradient Boosting. Their methodology had prediction accuracy of 93% and 91% on Cleveland and Framingham datasets

respectively [33]. On the same note, Ahmed et al. also applied feature selection techniques like Univariate and Relief to detect the useful attributes and trained a number of machine learning classifiers like Decision Tree, random forest and SVM to create an online heart disease prediction system [2]. Several studies have also showed comparative evaluations of different machine learning algorithms. Latha and Jeeva analyzed a number of classifiers such as Naive Bayes, BayesNet, Multilayer Perceptron and C4.5 decision tree. They reported that multi-classifier approaches could attain higher prediction accuracy than individual classifiers[27]. Yet, another investigation of Vayadande et al. compared machine learning models and deep learning models. They performed the experiments with the Cleveland heart disease data. Their findings verified competitive performance across several machine learning algorithms. XGBoost, Logistic Regression and Random Forest attained a final accuracy of 88.52. The results outperformed certain deep learning models in the same study [51]. With the recent progress in cardiovascular risk prediction, an increasing number has integrated machine learning with imbalance-aware learning strategies to achieve better predictiveness in skewed healthcare datasets. Recently, an increasing number of cardiovascular risk prediction studies have integrated machine learning with the use of imbalance-aware learning strategies to enhance the level of predictiveness in skewed healthcare datasets. Similarly, other hybrid oversampling/under-sampling, and different ensemble learning algorithms have shown greater robustness and/or minority-class recognition, thus favoring the incorporation of higher-balanced algorithms for medical prediction models [39, 40].While these studies show the potential use of machine learning in cardiac prediction, they predominantly use clinical data, lack adequate interpretability, and present a need for integrated ML-XAI frameworks that also integrates environmental determinants.

2.3 Explainable AI in CVD risk prediction

Explainable Artificial Intelligence (XAI) has enhanced the transparency and interpretability of machine learning models used for cardiovascular disease (CVD) risk prediction. The traditional machine learning models tend to be black boxes. Consequently, clinicians often find it difficult to understand how the models come up with their predictions. XAI techniques help explain model decisions and improve trust in AI-driven healthcare systems. [42]. Some of the studies investigated XAI to enhance interpretability of cardiovascular risk prediction models. Mridha et al. proposed an explainable machine learning framework for cardiovascular disease detection. Their study demonstrated that interpretable models can help clinicians understand the contribution of individual risk factors to prediction outcomes [35]. Shimizu et al. aimed at creating a machine learning system that forecasts major cardiovascular events. They had global and local interpretability methods in the system to enhance the reliability of models [44]. Recent studies have also emphasized the importance of explanation techniques such as SHAP for interpreting complex models. Moreno-Sanchez proposed an explainable AI model of predicting the survival of heart failure. Their model identified critical clinical variables that were strongly related to patient survival outcomes [34]. Athanasiou et al. applied explainable XGBoost model to predict cardiovascular risk. The patients on whom they studied were patients with type−2 diabetes. The explainability framework improved the interpretation of feature importance, thereby supporting clinical decision-making [8]. Other researchers have aimed at creating specific XAI models in predicting cardiovascular disease.

Table 1 Comparative analysis of related work on CVD risk prediction and mitigation

Ref	Application focus	Main goal	Method category	Data balancing technique	XAI integration	Linitations	How this study addresses it
[31]	Model interpretability	Evaluating Unified Models	SHAP	No	Yes	Not Applied to CVD Dataset directly	Applies SHAP to explain CVD risk
[38]	Interpretable ML	Explaining Black-Box Model	LIME	No	Yes	Focuses only on explanation method	Uses LIME to describe instance level prediction
[25]	CVD prediction	ML techniques evaluating risk assessment	RF, SVM, LR	No	No	Lacking in ML model prediction transperancy	Adding XAI to make prediction interpretable
[16]	CVD dataset evaluation	Assessing ML modeling on CVD dataset	ML Claasifiers	No	No	Focuses only on evaluation metrics	Adding XAI to interpret the output
[43]	CVD prediction	Improving Prediction analysis with ensemble learning	GBM, CatBoost, Neural Networks, SHAP	Not specified	Yes	Focuses only on predictive analysis	Using hybrid methods (RO, SM, TL) to improve minority class balancing
[28]	CVD risk prediction	Identifying risk factors of CVD	XGBC, SHAP	No	Yes	Focuses only feature modeling, not on predictive analysis	Combines both feature interpretation and prediction analysis
[11]	Heart disease prediction	XAI+ML based predictive System	ML, SHAP, LIME	Not specified	Yes	Limited analysis across datasets	Uses multiple datasets combined to interpret model accuracy
[47]	Heart disease prediction	Analysis of more developed ML interpretable model	Hybrid ML framework, SHAP, LIME	SM	Yes	Focuses on Detection	Explainability available more to assess risk prediction
[41]	Early CVD detection	Improving predictive accuracy with advanced boosting model	CatBoost, SHAP, LIME	No	Yes	Limitations in class balancing	Adding hybrid balancing techniques
[23]	CVD prediction	More Improving model than existing ones	CatBoost, GBC, ETC, SHAP, LIME, PIA	SM	Yes	Limitations in interpretability in XAI methods	Providing comparative analysis of XAI methods

Table 1 (continued)

Ref	Application focus	Main goal	Method category	Data balancing technique	XAI integration	Linitations	How this study addresses it
[1]	Heart disease prediction	Developing more interpretable model using Ensemble Modeling	Ensemble Model, SHAP, LIME	Not specified	Yes	Limitation in datasets	Multimodal dataset integration and hybrid data balancing methods
This Study	**CVD risk prediction**	**Developing accurate ML modeling with interpretable XAI methods**	**Hybrid ensemble ML models, SHAP, LIME, PIA**	**SM, RO, RO+TL, SM+TL, RO+IHT, SM+IHT, IHT**	**Yes**	**Addressing certain limitations of previous studies**	**Improving prediction analysis with accuracy and comparative analysis of XAI to get precise clinical decision of risk assessment**

The bold values in Table 1 represent the proposed contribution of this study compared with previous research works

Guleria et al. proposed a classification-based explainable framework for cardiovascular disease prediction. Their model increased transparency and clinicians were able to comprehend valuable contributory factors [19]. Sultan et al. proposed an ensemble model of stacking and explainable AI. Their model enhanced more accurate predictions and the decision-making process interpretation [46]. SHAP-based methods are widely used to classify important cardiovascular risk factors. These factors commonly include body mass index, hypertension, and cholesterol levels. SHAP provides both global explanations and local interpretations for individual predictions. Such descriptions allow clinicians to check the results of their models and learn more about medical risk factors [8, 34]. In general, combining XAI with machine learning allows predicting cardiovascular diseases with reliable and interpretable clinical results. Nevertheless, there are still a number of issues in the current research studies. Many previous studies have focused on only a limited number of environmental variables or specific diseases. These studies are not able to estimate the complex interactions of environmental exposure to population health. Traditional healthcare data also do not have high-resolution environment data both spatially and through time. This weakness minimizes the option of including environmental details in healthcare prediction models. To address this limitation, recent studies have adopted multimodal datasets that integrate health and environmental information. These datasets have satellite observations and other environmental variables. These datasets allow researchers to examine the associations between environmental factors and disease patterns. They also support the development of more inclusive predictive healthcare frameworks. More recent works also merge machine learning models with environmental risk prediction methods. Moreover, integrating explainable AI with machine learning can further enhance the predictive performance and interpretability of cardiovascular disease prediction models.

Table 1 presents a comparative analysis of previous studies on cardiovascular disease risk prediction. These studies primarily use machine learning and explainable artificial

intelligence in these studies. The comparison reflects application domains, learning methods, data balancing strategies, and explainability techniques. It also brings out the limitations that have been indicated in the prior research studies. The majority of the past work was geared towards prediction models based on conventional machine learning algorithms. Commonly adopted algorithms include Support Vector Machine, Random Forest, and Logistic Regression. These models have enhanced the accuracy of prediction in numerous studies that involve cardiovascular diseases. However, many of these models lack sufficient interpretability for clinicians and policymakers. Thus, prediction generation could be quite hard to comprehend in clinical practice. To address this concern, a number of researches proposed explainable AI practices. SHAP and LIME are among the most widely used explainability methods. These methods describe the contribution of features and enhance visibility of machine learning models. Recent studies have also discovered ensemble learning and hybrid machine learning models. These strategies have further enhanced the predictive capability of cardiovascular disease prediction systems. Several studies have integrated machine learning algorithms with explainable AI techniques. The approaches are useful in making sense of the forecasted cardiovascular risks and to define conceivable risk factors. Nonetheless, the majority of present research mostly is based on the clinical data. Prediction models usually do not include environmental and regional factors that have been associated with cardiovascular disease risk. The other weakness concerns the issue of class imbalance in healthcare data. It is possible that imbalanced data will lead to a prediction model that gives more weight to the majority. This may lead to biased predictions in healthcare applications. Research efforts sought to deal with this problem through oversampling. The Synthetic Minority Oversampling Technique (SMOTE) is one of the most widely approved approaches. Regardless of these attempts, there are a lot of studies that still work with small datasets. In their assessments, they mostly pay attention to prediction, as opposed to interpretability analysis. Another significant limitation is the limited combination of environmental and spatial health determinants. These are considerations that are seldom taken into account on cardiovascular disease prediction modeling. Moreover, many studies rely on a single explainability technique. They hardly compare several explainable AI methods that will enable improved interpretation of the model. To address these limitations, this study suggests a hybrid explainable machine learning framework. This structure is referred to as the HXAI-ML model. This model combines both environmental and epidemiological data based on SATHealth data. The proposed framework integrates ensemble machine learning algorithms with advanced data balancing techniques. These balancing techniques comprise Random Oversampling, SMOTE, Tomek Links and Instance Hardness Threshold. There are also a number of explainable AI methods used in the model. These are SHAP, LIME, and Permutation Importance Analysis techniques. These techniques provide both global and local explanations of prediction outcomes The proposed framework provides a more complete strategy for cardiovascular risk prediction. It combines environmental variables and machine learning and explainable AI techniques.

Machine learning has already shown its ability to enhance cardiovascular risk prediction. However, there remain a number of significant limitations. The well-established models are based on clinical data at the individual level and very few include environmental or satellite variables, although there is increasing evidence showing that climate, land use and environmental exposures have an impact on cardiovascular outcomes.

Second, many of the top-performing ensemble learning models are considered black boxes and would be unsuitable in the realm of health care decision-making and public health. Third, many studies investigating explainable AI have admittedly used just a single explanation approach, hindering the validation of explanations from various angles. Finally, few studies have tackled the problems of predictive accuracy and class imbalance together in a single framework with interpretation in mind. The gaps in these studies call for developing the proposed Hybrid Explainable Artificial Intelligence and Machine Learning (HXAI-ML) system which entails the eminence of integrating environmental-health data in a transparent manner. The system adopts advanced resampling methods, selecting a suite of ensemble learning and complementary Explainable AI (XAI) techniques to predict cardiovascular risk levels at the area-level.

3 Proposed methodology

This paper proposes a Hybrid Explainable Artificial Intelligence and Machine Learning (HXAI-ML) model to predict cardiovascular risks at the area level by integrating the environmental, satellite, and health record data. The framework unites ensemble machine learning models with the multi-level explainability methods that provide predictive robustness and forecastability. The overall workflow comprises 6 sequential instances: (i) dataset acquisition, (ii) data preprocessing and feature engineering, (iii) data balancing techniques, (iv) ensemble model training, (v) performance matrix investigation, (vi) explainability using different interpretable methods of explainable AI (XAI). The detailed chronological methodology is described in Fig. 1. This methodology is intended to predict performance with transparency and methodologically rigorous for highly impactful epidemiological modeling.

3.1 Dataset acquisition

The dataset used here was constructed by integrating heterogeneous data sources on the Core-Based Statistical Area (CBSA) level from Sathealth dataset. It involves the environmental and satellite-based variables, including the air quality, weather conditions, land surface attributes, and water-related features. The aggregation of data was done by area-weighted averages and annual aggregation was performed. Three-year windows

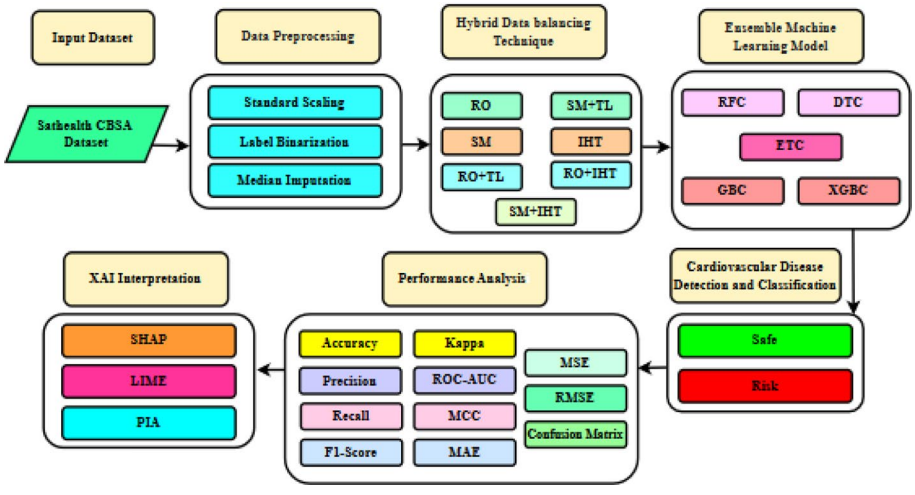


Fig. 1 Proposed hybrid XAI-ML approach for CVD detection

were used with the average of rolling three-year windows to form three-year lagged variables to capture effects of environmental exposure. From the Ohio dataset, the code (ICD-099) was taken which is for CVDs. During preprocessing, records with incomplete geographic identifiers or missing environmental or health information were omitted to confirm data consistency. Thus, the final dataset is 329 CBSA-year records after the cleaning and validation operation with 80 attributes.

3.2 Data preprocessing and feature engineering

Labelling was binarized, with cardiovascular risk of area (or region)-specific thresholds turning Safe (0) and Risk (1). The median was used to impute missing values of the satellite. The models were tested on a test dataset according to accuracy, precision and recall, F1-score, ROC-AUC, Cohen-Kappa, and MCC, and thresholds were set to trade-off sensitivity and specificity. Crucially, we trained the imputation model only on the training dataset, then applied it to the test dataset to avoid any data leaking across the line. Eighty of the environmental and health-related variables from the integrated SatHealth and Ohio ICD datasets were included in the initial feature set. To retain possible relevant environmental information and enable the influence of the most important predictors to be found by the explainable AI methods, all available variables were not eliminated by using aggressive dimensionality reduction, but were retained instead. Pearson correlation analysis was performed, with one feature from each highly correlated feature pair ($|r|>0.95$) removed to remove some features that would cause redundancy and multicollinearity. That is a conservative cut, to reduce near-redundant data while retaining variables that might represent different environmental qualities, while avoiding destabilizing the model and making it more opaque than desirable. After dividing the datasets into training and test sets by 80:20 ratio, we standardized all predictors with a Standard Scaler so every feature stayed on the same scale and the test dataset remained untouched. Furthermore, controlled hyperparameters were used for ensemble modelling to enhance the generalization performance.

3.3 Median imputation

The median imputation method is often used for numerical columns. It is preferred over the mean method because of the skewing of the data and less affected by extreme values. It is also used for imputing the missing values in the column to get continuous assessment of the dataset.

$$t_{ij}^{\text{imp}} = \begin{cases} t_{ij}, & \text{ifobserved} \\ \text{median}\left(t_j^{\text{train}}\right), & \text{ifmissing} \end{cases}$$

This prevents unnecessary data leakage.

3.4 Standard scaling

Standard Scaler is a method that employs Standard Normal Distribution (SND) and is utilized to scale the values of numeric feature. It converts data in such a way that the average will be 0 and the standard deviation will be 1. It calculates the difference between the mean of data and standard deviation. This makes the data driven towards zero and equalizes the variability.

$$m_{ij}^{scaled} = \frac{m_{ij} - n_j}{\sigma_j} \tag{1}$$

where n_j and σ_j are training-set mean and standard deviation.

To tackle the class imbalance problem, multiple balancing methods were used for both classes: Safe and Risk classes, such as Random Oversampling (RO), SMOTE, Random Oversampling & TL (RO+TL), SMOTE & TL (SMOTE+TL), Instance Hardness Threshold (IHT), Random Oversampling & IHT (RO+IHT) and SMOTE & IHT (SMOTE+IHT). After the split into train-test, all balancing procedures were done only on the train part and the test part was left with the same class distribution as the original. This protocol avoided the leakage of information; it enabled to compare the various resampling strategies under similar conditions of the experiments. However, since there was no attempt here to optimize the model's predictive power, the feature selection was based instead on redundancy reduction that would enable evaluation of the relative importance of the variables selected in the subsequent explainable AI very analysis.

3.5 Cross-validation strategy

A 5-fold stratified cross-validation experiment was performed on the best performing model to further demonstrate its robustness and generality. To avoid information leakage, the same preprocessing steps as described above were independently performed within each fold. Which are involving median imputation, correlation-based feature filtering, data balancing, standardization, isotonic calibration, and threshold optimization. Mean and standard deviation for each of the five stratified folds indicates the spread and stability of the proposed framework. Repeated evaluation further minimizes the chance the performance demonstration in the prediction is due to a specific train-test partition, and offers further validation of the stability of proposed HXAI-ML model.

3.6 Data balancing with necessary techniques

In our dataset, in order to find class imbalance, we used a combination of oversampling and undersampling techniques like Random Oversampling (RO), Tomek Links (TL), SMOTE and Instance Hardness Threshold (IHT). These procedures influence the testing of both minor and major classes, confirming the model's training on balanced data. We examined the proposed methods:

3.7 Random oversampling (RO)

Random Oversampling is a method of dealing with data imbalance in a dataset by adding the number of samples in the minor class. It operates by randomly copying existing examples in the minority group, until the data sets are even or even all the closer. This is a straightforward method, which is easy to apply and it allows ML models to be more attentive to the minority class when training. Nevertheless, since it barely generates new information but only replicates the available ones, this could lead to the risk of overfitting, particularly when the minority group has a very small number of unique cases.

3.8 Synthetic minority oversampling technique (SM)

Our proposed approach is SM. SMOTE is another popular oversampling method that does not just replicate, but creates synthetic samples. It operates by choosing a sample

of a minority class and forming novel artificial values on the line segments in between it and its closest minority neighbors. This encourages the diversity of the minority group and eliminates the possibility of overfitting when compared to the simple random oversampling. The SMOTE enhances the learning capacity of the classifiers when dealing with unbalanced data sets however in some cases, it also adds artificial points that may give a clash to the majority class hence it may add noise. The synthetic sample creation is defined as:

$$a_{\text{new}} = a_{\min} + \lambda \cdot (a_{\text{neighbor}} - a_{\min}) \tag{2}$$

where,

a_{new}= synthetic instance,
$a_{\min}$ = minority class instance,
a_{neighbor} = randomly selected neighbor of $x_{\min}$,
λ = arbitrary digit between 0 and 1.

3.9 Random oversampling with Tomek links (RO+TL)

It is a hybrid method used to first balance the sample by sampling each majority class two or three times, and then uses Tomek Links to clean the sample. This step adds oversampling, which adds the examples of the minority classes and allows the model to learn more effectively with underrepresented classes. The Tomek Links technique then eliminates the duplicate sample pairs of contrasting classes, and this increases separation between classes. This combination assists in reducing the noise that may be caused by oversampling and may result in more confident classification. Let c_i and c_j be a PIA of examples fitting to dissimilar classes; they form a Tomek link if:

$$e(c_i, c_j) < e(c_i, c_k) \text{ and } e(c_j, c_i) < e(c_j, c_l) \tag{3}$$

where:

- $e(c_i, c_j)$ is the distance between c_i and c_j,
- c_k and c_l are instances from other classes.

3.10 SMOTE with Tomek links (SM+TL)

This is a combination that balances class by combining both oversampling and under sampling. The first is that SMOTE creates artificial samples related to the minority group in order to boost its numbers. Following this, the Tomek Links will be applied to eliminate the ambiguous samples or overlapping samples between the minority and the majority classes. A Tomek Link is two samples which are of different classes and are closest neighbors to each other. Eliminating them surpasses cleaning the boundary between the classes and decreases the noise to enhance the performance and the generalization capacity of the machine learning models.

3.11 Instance hardness threshold (IHT)

This sampling technique eliminates hard or noisy data of the majority sample due to its the predicted classification hardness. It predicts the chance that a given instance will be misclassified based on a machine learning model and it is called instance hardness. Ambiguous or noisy samples with large values of hardness are removed out of the

dataset. The approach assists in improving the performance of the models because it eliminates misleading samples and trains on the more understandable patterns in the data.

3.12 Random oversampling with instance hardness threshold (RO+IHT)

It is a combination of simple random duplication and noise reduction capability of an instance hardness filter. First, random oversampling, which is a variant of the above, raises the quantity of the samples of a minority class by copying the previous samples. After that, the Instance Hardness Threshold approach analyses the data and eliminates the highly mis classifiable samples. This process aids in keeping a dataset balanced and also boosts it quality by eliminating ambiguous or misleading data then can strengthen machine learning models.

3.13 SMOTE with instance hardness threshold (SM+IHT)

In this method, it targeted at improving the dataset and cleaning it at the same time. Under this method, SMOTE would initially be used to create synthetic minority samples and balance the dataset. Then, the Instance Hardness Threshold method is used to consider which instances are highly likely to be misclassified and eliminate them accounted to be too hard or noisy. This combination is beneficial because it results in a higher level of representation of the minority classes, as well as enhances the quality of the data by getting rid of problematic samples.

3.14 Proposed approach (SM)

Out of the tested approaches, SM exhibits the highest performance in the decrease of the imbalance in the classes and enhancement of the model generalization. FIGURE 2 illustrates the flow chart of the study framework we used to develop the predictive decision from the dataset.

3.15 Ensemble machine learning (EML) models

We boosted predictive stability and robustness by using several ensemble-based classifiers, each combining its models like a well-tuned orchestra adjusting to every note. The EML framework used five models: Decision Tree Classifier (DTC), Random Forest (RFC), Extra Trees (ETC), Gradient Boosting (GBC), and the sharper, faster Extreme Gradient Boosting (XGBC).

3.16 Decision tree classifier (DTC)

DTC is a tree-based model that creates a tree structure. With the tree structure, it split the data into branches based on features to predict the outputs. DTC is easily interpretable to understand and visualize. the Gini impurity at a node is:

$$F = 1 - \sum_{i=1}^{G} q_i^2 \tag{4}$$

where, q_i is the proportion of class i instances, and G is the total number of classes.

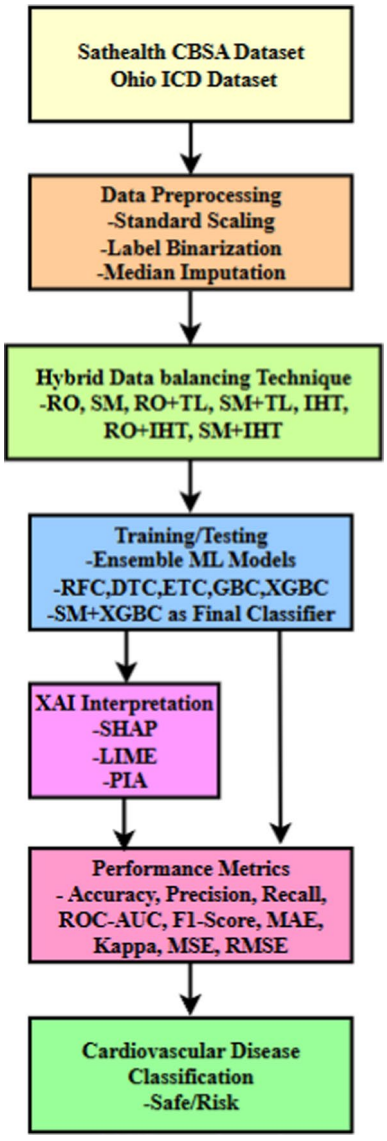


Fig. 2 Schematic flow diagram of the proposed HXAI-ML framework predicting CVD risk

3.17 Random forest classifier (RFC)

An ensemble method building multiple decision trees during the training period. Then they merge their predictions by averaging for regression or majority vote for classification. In this mechanism, each tree is trained on a random.

subset of features creating diversity. Mathematically, the final prediction $\hat{y}$ of RFC is given by:

$$\hat{z} = \frac{1}{S} \sum_{t=1}^{S} b_t\,(y)\,(5)$$

where $b_t(y)$ represents the prediction of the t-th tree and S is the total number of trees.

3.18 Extra trees classifier (ETC)

ETC is similar kind of RFC, which ensembles de-correlated decision trees. It generates threshold for each feature instead of finding the discriminative threshold for each feature. Then it preferences the top among the random splits. Also, it reduces variance and make the model faster than others.

3.19 Gradient boosting classifier (GBC)

Gradient Boosting Classifier is constructed in a series of trees with each succeeding tree seeking to correct the mistakes of the predecessor one. It is slightly slower model technique but gives higher accuracy. The disadvantage of the technique is the overfitting tendency because of the not tuning correctly. The update rule for the model prediction $\hat{R}(k)$ at iteration m is:

$$\hat{R}_x\left(k\right) = \hat{R}_{x-1}\left(k\right) + \eta \cdot h_m\left(k\right), (6)$$

where, η is the learning rate and $h_m(k)$ is the m-th base learner.

3.20 Extreme gradient boosting classifier (XGBC)

XGBC is an advanced and optimized version of GBC. It enhances both the speed and performance. It offers good controlling over overfitting by initializing regularization. The regularized objective function is:

$$\mathcal{L}\left(\theta\right) = \sum_{i=1}^{n} m\left(n_i, \hat{y}_i\right) + \lambda \sum_{j=1}^{k} h_j^2 \quad (7)$$

where the loss function is denoted by m, the expected amount is $\hat{y}_i$, the system parameters are h_j, and the term used to regularize is λ.

We chose these models because they can pick up nonlinear patterns, manage overlapping variables, and model complex feature interactions often found in environmental epidemiology. Among these methods, XGBoost stood out for its regularization, shrinkage tuning, and gradient boosting framework improve generalization performance while reducing overfitting. We evaluated the model on a separate test set, measuring how well it performed using accuracy (total number of correct predictions), precision (number of predictions that are accurate divided by the total number of predictions), recall or sensitivity (number of accurate predictions relative to the overall number of predictions), F1-score (trade-off between precision and recall), ROC-AUC (capacity to differentiate classes), Cohen's Kappa, and the Matthews Correlation Coefficient (MCC). We fine-tuned the thresholds where needed to keep sensitivity and specificity in balance.

3.21 Explainable artificial intelligence (XAI) techniques

To keep predictions clear and easy to interpret, the HXAI-ML framework blends three explainability tools-SHAP, LIME, and Permutation Importance Analysis-working together like lenses that reveal how each feature shapes an outcome.

3.22 Shapley additive explanation (SHAP)

This method is a widely adopted explainability technique of explaining machine learning models. This algorithm is utilized to know how the data features or variables make

a contribution to model predictions. It is a game-theoretic method of modeling the outputs of machine learning. The role of each feature is measured by Shapley values, which show its effect on the prediction of the model.

3.23 Local interpretable model-agnostic explanation (LIME)

LIME is one of the prominent XAI frameworks provided for visualizing and interpreting ML prediction approach. It aims to produce the interpretation of each individual prediction of how it approaches to the direction of the predictions. It is a local interpretation like the global model behavior. It is useful for the prediction of complicated models at a local level.

3.24 Permutation importance analysis (PIA)

PIA is a naive though useful technique of identifying the importance of features, where the importance of one feature is derived as the result of permutation of all features. It is used to bring to the fore the most significant aspects most easily, which is very essential in medical decision making. PIA has less computer power requirements relative to alternative methods (SHAP and LIME). It works directly with the performance measures of accuracy and the F1 score thus it offers simple to understand results.

The analysis was done by SHAP to understand the contribution of each feature to the model output. LIME provided local explanations for individual CBSA instances by approximating the complex model with a simpler interpretable model, thereby identifying the factors contributing to each prediction. Another tool named Permutation Importance Analysis (PIA) was used that reveal the level of significance of a feature by observing how the accuracy of the model is altered by changing the values of a given feature. These methods provide intuitive, model-independent explanations and are generally less sensitive to biases associated with traditional impurity or gain-based importance measures. We have picked the 17 most important features out of 80 and taken a closer look at them based on their mean absolute SHAP values and their permutation importance value scores, to ensure that they indicate actual predictive power with not merely random chance. A combination of explainability methods and ensemble learning addresses a significant issue with environmental health modelling: how to make good predictions alongside understanding why. Ensemble models are more robust in their predictions and are usually black boxes. The suggested HXAI-ML framework addresses this limitation by integrating ensemble models with transparent and theoretically grounded explanation techniques. Furthermore, the hybrid framework is reproducible, and its interpretable outputs improve its potential utility for public health planning and environmental risk monitoring.

4 Performance analysis

4.1 Experimental setup

Table 2 highlights the features of the dataset, preprocessing, machine learning algorithms and evaluation measures in this cardiovascular risk prediction study on CBSA level. It further provides the list of the software libraries and hardware configuration to derive a reproducible experimental set-up.

The dataset was split into training and testing sets, as previously stated with an 80:20 ratio, to maintain the same distribution of classes in each set. Random numbers were

Table 2 System specifications and experimental framework

Component	Description
Dataset characteristics	The research involves the use of a cardiovascular risk data at CBSA-level which can be described as 329 observations and 80 environmental and contextual attributes based on built-in satellite and environmental indicators. In the form of a CBSA year of observation, each observation is a data on environmental exposure conditions and local cardiovascular risk. The problem is stated as a binary classification problem with areas defined as either Safe (0) or Risk (1) depending on cardiovascular risk levels
Data preprocessing pipeline	Before the development of models, some preprocessing functions had been performed to provide data consistency and statistical validity. The gaps in values between satellites retrieved or missing environmental records were completed by the median imputation as fitted to the training set. Standard scaling was used to normalize the feature distributions capturing the difference in the scales across variables. Label binarization was employed to convert the target variable into numerical format and generate three-year lagged environmental variables that reflect cumulative exposures effects that determine the contribution of cardiovascular risks
Training and testing	In order to test the performance of the generalization of the model, the training and testing split was performed with 80% divided into training and 20% on testing as well. Model learning and hyperparameter setup were carried out on training subset and independent testing on predictive performance was carried out on testing subset. Such division helps to guarantee the impartial evaluation of the model or its ability to solve methods on hidden CBSA observations
Class balancing techniques	Since the dataset had a skew or unequal distribution of classes in both Safe and Risk regions, a number of data balancing methods were used when training the model. These are Random Oversampling (RO), Synthetic Minority Over-sampling Technique (SMOTE), Random Oversampling with Tomek Links (RO+TL), SMOTE with Tomek Links (SM+TL), Instance Hardness Threshold (IHT), RO+IHT and SMOTE+IHT. These methods enhance the representation of minority classes and make the models acquire more credible limits of decisions
Predictive algorithms	Predication of cardiovascular risks was done using five supervised ensemble machine learning classifiers. These are the Random Forest Classifier (RFC), Decision Tree Classifier (DTC), Extra Trees Classifier (ETC), Gradient Boosting Classifier (GBC) and Extreme Gradient Boosting Classifier (XGBC). The rationale associated with the choice of these algorithms was the ability of multivariate environment data handling, nonlinear relationship capture, and also their ability to deliver a high level of predictive accuracy in multivariate classification problems
XAI implementation	Three explainable artificial intelligence methods were used in order to be able to interpret model predictions. SHAP (SHapley Additive exPlanations) provided relative and absolute contribution of features both globally and locally on game-theoretic principles. LIME (Local Interpretable Model-agnostic Explanations) offered instance-level explanations via an approximation of the model locally as a surrogate model that is easy to interpret. PIA was used to measure the importance of global feature by permuting features randomly and using the evaluation of the impact of the same in the model performance
Performance metrics evaluation	A notable evaluation metrics comprised of Accuracy, Precision, Recall, F1-score, ROC-AUC, Cohen's Kappa, Matthews Correlation Coefficient (MCC), Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root Mean Squared error (RMSE) was used to evaluate the performance of the model. Later 5-fold cross-validation on the best performing model was used to compare with the hold-out test. To determine class-wise predictive ability, the additional assessment was carried out with the help of confusion matrices and classification reports
Tools and libraries used to compute	Python-based libraries of scientific computing were used to implement the experimental framework. The model was developed, preprocessed, and evaluated using scikit-learn, numerical computation and data manipulation were done using NumPy and Pandas, visualization was done using Matplotlib and Seaborn, and class balancing techniques such as SMOTE or Tomek links were performed using imbalanced-learn
Best performing model	According to the comparative experimental results, the best predictive performance was found in the use of SMOTE + Extreme Gradient Boosting (SM+XGBC) configuration, which was the most accurate and the best recall and F1-score when it came to predicting high-risk CBSA regions

always generated from the same seed (42) which facilitated the repeatability of the experiments. Data imputation was done via median and highly correlated features were filtered out using correlation value. To overcome this phenomenon, SMOTE was used only on the training data and not the testing data (information leakage would not be allowed). All the training data have been balanced and standardized prior to the creation of the model.

Empirically selected parameters for the XGBoost's classifier were the number of estimators (600), max tree depth (3), learning rate (0.05), subsampling rate (0.8), and column sampling rate (0.7). Isotonic regression (CalibratedClassifierCV) was used to perform probability calibration. The threshold tuning was also done by computing the F1 score for prediction on the validation set for various thresholds (0.30 to 0.70) and using the threshold that gives the highest F1 score.

4.1.1 Data collection

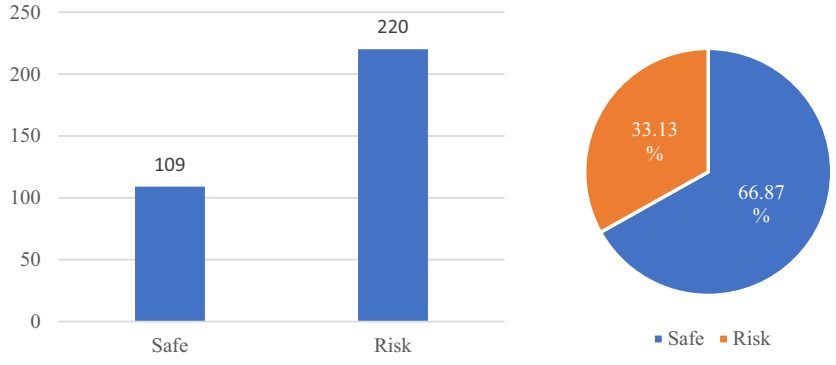
A combined CBSA-level dataset was also developed by joining the regional cardiovascular disease risk data and environmental indicators obtained via satellites. The further analytical data set consists of 329 observations (CBSA-year) of either Safe or Risk (0 or 1).

4.1.2 Dataset characteristics and class distribution

With 109 Safe instances and 220 Risk cases, the preprocessed dataset presented a substantial class imbalance. Both count-based and percentage-based visualizations of the original labels are shown in Fig. 3 to illustrate the distribution. The imbalance draws attention to the dataset's high-risk CBSA multitude and highlights the necessity of suitable resampling techniques to avoid biased model learning. Synthetic Minority Oversampling Technique (SMOTE) was used exclusively on the training data to balance the classes.

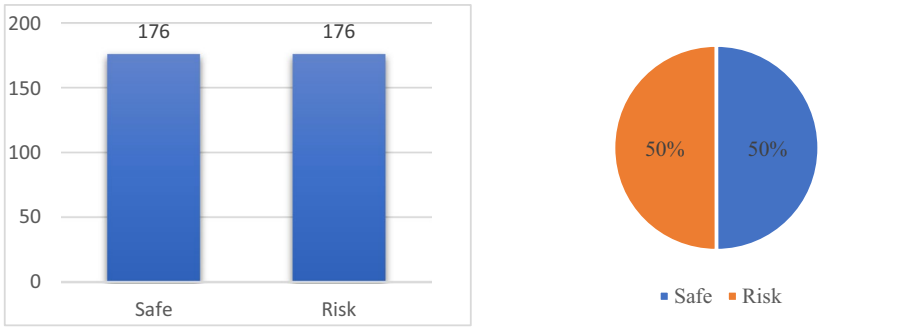
As seen in Fig. 4, the class distribution was perfectly balanced after resampling, with 176 Safe and 176 Risk instances (50%–50%). Predictive fairness and constancy were boosted by the models' ability to learn decision boundaries without supporting the majority class thanks to this balanced distribution.

Table 3 contains automatic calculation of all the evaluation metrics in order to maintain consistency across the entire experimental pipeline to avoid manual transcription



(a) The Count of Each Class (b) The Percentage of Each Class

Fig. 3 Original data distribution of each class. **a** The count of each class **b** The percentage of each class



(a) The Count of Each Label (b) The Percentage of Each Label

Fig. 4 Class distribution after applying SM data balancing method. **a** The count of each label **b** The percentage of each label

errors. ROC-AUC was calculated using predicted probabilities, while Cohen's Kappa and MCC were calculated based on the last truth table label.

4.1.3 Data visualization

Important information about the dataset's class structure can be collected from the visualizations of the original and post-SMOTE distributions. To make sense of the results and choose the correct path to fix the data, firstly had to know why the groups were uneven.

4.1.4 Performance metrics

Accuracy, precision, recall, F1-score, Cohen's Kappa, Matthews Correlation Coefficient (MCC), ROC-AUC, mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE) are some of the common classification and error-based metrics that are used to assess the models' predictive power. These metrics offer a detailed evaluation of prediction error, agreement strength, balanced classification performance, enabling in-depth comparisons between various models and balancing techniques.

4.2 Analysis of results

Our experimental study evaluated the performance of the proposed SM +XGBC hybrid model. The model was assessed using the preprocessed CBSA level of cardiovascular disease prediction dataset. The framework integrated environmental and epidemiological data. The dataset was comprised of 329 CBSA year observations that had 80 features used to describe environmental and health conditions. The binary target variable classified regions as Safe (0) or Risk (1). The sample size combines 109 safe cases and 220 risk cases indicating a medium level of the imbalance between the classes. An 80:20 train-test split was used in data distribution with 80% of the data used for model training and remaining for testing of the model. Pearson correlation analysis was used to eliminate predictors that have a high correlation with an absolute correlation coefficient of more than 0.95. This move assisted to minimize multicollinearity of variables in data. A simple median imputation strategy was employed to handle missing values. Before model training, all characteristics were standardized using Standard Scaler. Several data balancing algorithms were used subsequently to solve class imbalance. Such algorithms were Random Oversampling, SMOTE, RO with Tomek Links and SMOTE with Tomek

Table 3 Performance analysis of ML models on proposed CBSA-level environmental and cardiovascular risk dataset

Exp no	Data balancing technique	Model	Accuracy	Precision	Recall	F1score	Kappa	ROC-AUC	MCC	MAE	MSE	RMSE
1	Without											
		DTC	78.19	86.55	80.47	83.40	51.74	84.63	52.03	21.81	21.81	46.70
		RFC	86.70	94.02	85.94	89.80	70.82	95.46	71.41	13.30	13.30	36.47
		ETC	85.64	91.74	86.72	89.16	67.95	95.43	68.18	14.36	14.36	37.90
		GBC	86.70	92.56	87.50	89.96	70.32	93.65	70.56	13.30	13.30	36.47
		XGBC	84.57	92.31	84.38	88.16	66.15	92.37	66.71	15.43	15.43	39.28
2	RO	DTC	84.85	92.50	84.09	88.10	67.39	88.69	67.97	15.15	15.15	38.92
		RFC	92.42	95.35	93.18	94.25	83.15	93.49	83.19	7.58	7.58	27.52
		ETC	90.91	93.18	93.18	93.18	79.55	92.98	79.55	9.09	9.09	30.15
		GBC	89.39	91.11	93.18	92.13	75.86	94.52	75.91	10.61	10.61	32.57
		XGBC	92.42	95.35	93.18	94.25	83.15	94.94	83.19	7.58	7.58	27.52
3	SM	DTC	89.89	93.02	90.91	91.95	76.40	87.45	76.45	10.61	10.61	32.57
		RFC	92.42	95.35	93.18	94.25	83.15	93.49	83.19	7.58	7.58	27.52
		ETC	89.39	89.36	95.45	92.31	75.29	92.36	75.72	10.61	10.61	32.57
		GBC	90.91	91.30	95.45	93.33	79.07	94.99	79.26	9.09	9.09	30.15
		XGBC	95.45	97.67	95.45	96.55	89.89	96.23	89.94	4.55	4.55	21.32
4	RO+TL	DTC	80.30	91.89	77.27	83.95	58.95	92.95	60.44	19.70	19.70	44.38
		RFC	90.91	93.18	93.18	93.18	79.55	92.98	79.55	9.09	9.09	30.15
		ETC	89.39	91.11	93.18	92.13	75.86	92.67	75.91	10.61	10.61	32.57
		GBC	90.91	95.24	90.91	93.02	80.00	94.89	80.18	9.09	9.09	30.15
		XGBC	93.94	97.62	93.18	95.35	86.67	95.76	86.86	6.06	6.06	24.62
5	SM+TL	DTC	86.36	92.68	86.36	89.41	70.33	89.67	70.68	13.64	13.64	36.93
		RFC	90.91	93.18	93.18	93.18	79.55	93.29	79.55	9.09	9.09	30.15
		ETC	90.91	91.30	95.45	93.33	79.07	92.87	79.07	9.09	9.09	30.15
		GBC	92.42	93.33	95.45	94.38	82.76	93.90	82.76	7.58	7.58	27.52
		XGBC	92.42	97.56	90.91	94.12	83.52	95.97	83.52	7.58	7.58	27.52

Table 3 (continued)

Exp no	Data balancing technique	Model	Accuracy	Precision	Recall	F1score	Kappa	ROC-AUC	MCC	MAE	MSE	RMSE
6	IHT	DTC	80.30	91.89	77.27	83.95	58.95	82.44	60.44	19.70	19.70	44.38
		RFC	89.39	89.36	95.45	92.31	75.29	92.25	75.72	10.61	10.61	32.57
		ETC	89.39	89.36	95.45	92.31	75.29	91.84	75.72	10.61	10.61	32.57
		GBC	84.85	97.22	79.55	87.50	68.75	92.15	71.00	15.15	15.15	38.92
		XGBC	83.33	7.14	77.27	86.08	65.98	94.73	68.69	16.67	16.67	40.82
7	RO+IHT	DTC	86.36	92.68	86.36	89.41	70.33	87.81	70.68	13.64	13.64	36.93
		RFC	90.91	93.18	93.18	93.18	79.55	92.98	79.55	9.09	9.09	30.15
		ETC	89.39	91.11	93.18	92.13	75.86	92.67	75.91	10.61	10.61	32.57
		GBC	90.91	95.24	90.91	93.02	80.00	95.56	80.18	9.09	9.09	30.15
		XGBC	92.42	97.56	90.91	94.12	83.52	95.66	83.93	7.58	7.58	27.52
8	SM+IHT	DTC	87.88	89.13	93.18	91.11	72.09	88.84	72.09	12.12	12.12	34.82
		RFC	90.91	93.18	93.18	93.18	79.55	92.87	79.55	9.09	9.09	30.15
		ETC	86.36	87.23	93.18	90.11	68.24	90.70	68.62	13.64	13.64	36.93
		GBC	89.39	93.02	90.91	91.95	76.40	93.49	76.40	10.61	10.61	32.57
		XGBC	90.91	95.24	90.91	93.02	80.00	96.18	80.18	9.09	9.09	30.15

links. Additional techniques included Instance Hardness Threshold, RO+IHT and SMOTE+IHT. A series of machine learning classifiers was trained on these datasets that had been balanced. The evaluated classifiers were the Random Forests, Decision Tree, Gradient Boosting, XGBoost and Extra Trees classifiers. The F1-score was selected as the primary optimization metric because it offers a balanced evaluation of precision and recall, particularly for imbalanced datasets. The best classification balance was sought between threshold values that were searched between 0.30 and 0.70. The multiple evaluation metrics were used to evaluate the model performance. These were accuracy, precision, recall, F1-score, ROC-AUC, kappa, MCC, MAE, MSE and RMSE.

The experimental results showed that SMOTE accomplished the strongest overall performance among the evaluated balancing techniques. The best results were realized with the use of the XGBC model plus the SM as shown in Table 3. The model had an accuracy of 95.45, a precision of 97.67 and an F1-score of 96.55. Two factors may have contributed to the strong performance of the XGBoost model. First, the gradient boosting optimization algorithm enhances learning on complicated patterns within the data. Second, SMOTE develops synthetic minority samples that allow one to balance the dataset effectively. These synthetic samples assist the model to better signify the decision boundaries of the minority class. Consequently, the model enhances the predictive accuracy and generalization capability.

In contrast, the DTC method trained without data balancing technique shows the lowest result in terms of accuracy, precision, recall and all other parameters. This may be attributed to its higher variance sensitivity and limited representation capability. Decision tress intended to overfit and sensitive to class imbalance. When class imbalance is not properly omitted, the model gives priority to the majority class and splitting towards them.

Random Forest and Gradient Boosting achieved intermediate performance without data balancing. This may be because their ensemble mechanisms alone cannot fully recompense for class imbalance. By generating synthetic minority samples, the model was better able to capture the associations between environmental factors and cardiovascular risk. As a result, it executed better than any other model tested.

4.2.1 Comparative performance analysis

The overall comparative analysis of the models indicates SMOTE was the most consistent performed model across the classifiers used. The hybrid SM+XGBC model achieved the best performance. Models such as GBC, ETC and RFC demonstrated satisfactory improvements when combined with techniques like RO and SMOTE+TL. The superior kappa, MCC and F1-score achieved by the SMOTE+XGBC model, ensures that this method efficiently withdraw class imbalance. Furthermore, the model effectively captures complex nonlinear relationships among the environmental variables. These findings highlight the importance of merging advanced data balancing strategies with robust learning algorithms to improve predictive performance in regional environmental health risk modeling.

4.2.2 Analysis of the top-performing model

In the experiments shown that the CBSA-level environmental oriented cardiovascular risk dataset, the best-performing model across all evaluated configurations was

illustrated in Table 4. The system that employed SMOTE together with XGBC worked best. This framework achieved the highest accuracy of 95.45%, along with greater precision (97.67%), recall (95.45%), and F1-score (96.55%). Strong agreement metrics, such as kappa reaching 89.89% and MCC reaching 89.94%, respectively indicate a strong degree of consistency on safe and risk classification. Additionally, the model achieved the lowest error measures among all evaluated configurations, with an RMSE of 21.32%, MAE and MSE values of 4.55%. These values indicate strong generalization ability on the independent test set. The SM + XGBC configuration continuously showed better stability and discrimination power when compared to other balancing strategies like RO, RO+TL, SM+TL, and IHT-based techniques. The model represented the nonlinear relationships between environmental factors and local cardiovascular risk. This performance was likely improved by the combination of gradient boosting and synthetic minority oversampling. Overall, these results prove the value of merging advanced resampling strategies with robust ensemble learning techniques for accurate and scalable area-level health risk prediction.

4.2.3 Cross-validation analysis

To investigate the stability of the proposed model, a 5-fold stratified cross validation was used on the best-performing model to assess its robustness and quantify the variability of predictive performance. as stated in Table 5. The SMOTE+XGBoost framework achieved an average accuracy of 93.31 ± 2.26%, an average ROC-AUC of 95.24 ± 1.48%, and an average F1-score of 95.03 ± 1.73%. The standard deviations of the values recorded across the folds are relatively low, which indicates the model has a consistent predictive capacity and performance fluctuations are relatively small on different subsets of the learning data.

4.2.4 Comparison with baseline model

Logistic Regression is added as the conventional baseline along with other classifiers; Random Forest and Extreme Gradient Boosting Classifier as stated in Table 6 to further assess the proposed framework. The logistic regression model showed the highest ROC-AUC value (98.70%), doing a good job of class discrimination among different decision levels. An ensemble learning model was more accurate than it, and its values of accuracy (87.69%), F1 score (90.24%), Cohen's Kappa (73.82%) and Matthews correlation coefficient (75.28%) were better. Random Forest provided better overall classification performance, while XGBC (95.45%) had the highest accuracy, F1 score (96.55%), Kappa score (89.89%), and MCC score (89.94%). The result showed Logistic Regression had excellent discriminative performance but the proposed SMOTE+XGBC framework yielded a better balance of discrimination, classification accuracy, and prediction reliability.

4.2.5 Classification report analysis

The classification report analysis demonstrates the efficiency of the SMOTE-based data balancing strategy in combination with various ensemble learning models. As shown in Table 7, The models' performance differs between the classes and the XGBC classifier continuously exhibits better class-wise discrimination. For the Safe class, XGBC attained a precision of 91.30%, recall of 95.45%, and an F1-score of 93.33%. Hence XGBC proved to have a strong ability to accurately identify safe classes while reducing false negatives.

Table 4 Top-performing models found in every single experiment on the dataset

Exp no	Data balancing technique	Model	Accuracy	Precision	Recall	F1-score	Kappa	ROC-AUC	MCC	MAE	MSE	RMSE
1	Without	RFC	86.70	94.02	85.94	89.80	70.82	95.46	71.41	13.30	13.30	36.47
2	RO	XGBC	92.42	95.35	93.18	94.25	83.15	94.94	83.19	7.58	7.58	27.52
3	**SM**	**XGBC**	**95.45**	**97.67**	**95.45**	**96.55**	**89.89**	**96.23**	**89.94**	**4.55**	**4.55**	**21.32**
4	RO+TL	XGBC	93.94	97.62	93.18	95.35	86.67	95.76	86.86	6.06	6.06	24.62
5	SM+TL	XGBC	92.42	97.56	90.91	94.12	83.52	95.97	83.52	7.58	7.58	27.52
6	IHT	RFC	89.39	89.36	95.45	92.31	75.29	92.25	75.72	10.61	10.61	32.57
7	RO+IHT	XGBC	92.42	97.56	90.91	94.12	83.52	95.66	83.93	7.58	7.58	27.52
8	SM+IHT	XGBC	90.91	95.24	90.91	93.02	80.00	96.18	80.18	9.09	9.09	30.15

The bold values in Table 4 indicate the best-performing model configuration among all evaluated experiments

Table 5 Cross-validation of the best performing model

Metric	Hold-out test	5-Fold CV
Accuracy	95.45	93.31 ± 2.26
ROC-AUC	96.23	95.24 ± 1.48
F1-Score	96.55	95.03 ± 1.73

Table 6 Comparison between simple baseline like logistic regression with the best-performing model (SMOTE)

Exp no	Model	Accuracy	Precision	Recall	F1-score	Kappa	ROC-AUC	MCC	MAE	MSE	RMSE
1	LR	87.69	97.37	84.09	90.24	73.82	98.70	75.28	12.31	12.31	35.08
2	RFC	92.42	95.35	93.18	94.25	83.15	94.94	83.19	7.58	7.58	27.52
3	XGBC	**95.45**	**97.67**	**95.45**	**96.55**	**89.89**	**96.23**	**89.94**	**4.55**	**4.55**	**21.32**

Table 7 Classification report of ML model using SM on the dataset for CVD risk prediction

Model	Class	Precision	Recall	F1-score
DTC	Safe	82.61	86.36	84.44
	Risk	93.02	90.91	91.95
RFC	Safe	85.71	81.82	83.72
	Risk	91.11	93.18	92.13
ETC	Safe	67.86	86.36	76.00
	Risk	92.11	79.55	85.37
GBC	Safe	80.00	90.91	85.11
	Risk	95.12	88.64	91.76
XGBC	**Safe**	**91.30**	**95.45**	**93.33**
	Risk	**97.67**	**95.45**	**96.55**

The bold values in Table 7 highlight the superior performance of the proposed SMOTE + XGBC model compared with the baseline Logistic Regression and Random Forest models

The model performed even better for the risk class at a higher precision (97.67%), recall (95.45%), and F1-score (96.55%), which is linked with high quality control of the false positives and detection of high-risk regions. ETC produced relatively less accuracy with safe category and still have decent recall. DTC and RFC proved competitive but lower stability in comparison to other classifiers. In addition, the performance of GBC was also highly deviant and its' total scores were still inferior to those obtained by XGBC. Overall, SMOTE combined with XGBC has produced balanced performance in both classes significantly. The high reliability and validity in distinguishing levels of risk of cardiovascular in different regions following environmental determinants are ensured by the high values of precision and recall used to determine safe and risk classes.

4.2.6 Complexity report analysis

The SMOTE+XGBC configuration attained strong predictive performance with low computational cost. According to Table 8, XGBC required 0.563 s for training and 0.004 s for predicting, which sums up a total of 0.567 s. Therefore, XGBC is appropriate in public health where false-negative is of great concern. In contrast, GBC achieved competitive predictive performance, but it had the highest computational timestamp, requiring 5.085 s for training and 5.091 s overall. With respective total execution times of 0.843 and 0.614 s, RFC and ETC exhibited moderate computational demands. DTC performed noticeably worse than other ensemble-based models, despite being the fastest overall with execution time 0.011 s in total.

Table 8 Complexity analysis of ML models

Data balancing technique	Model	Training time (sec)	Prediction time (sec)	Total time (sec)
SM	DTC	0.011	0.000	0.011
	RFC	0.817	0.026	0.843
	ETC	0.582	0.032	0.614
	GBC	5.085	0.006	5.091
	XGBC	**0.563**	**0.004**	**0.567**

The bold values in Table 8 indicate the computational efficiency of the proposed SMOTE + XGBC model

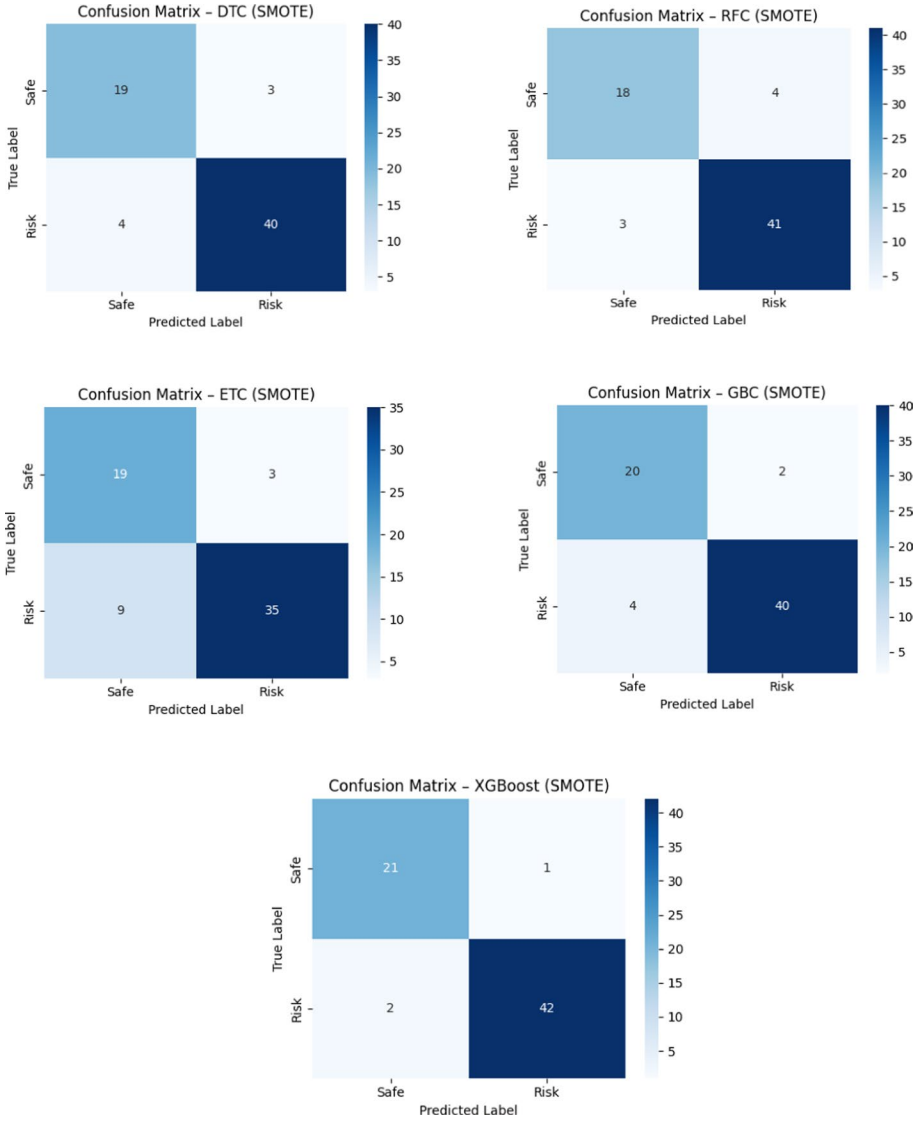


Fig. 5 Confusion matrix of ML modeling using SM on the dataset

4.2.7 Confusion matrix analysis

The confusion matrix analysis shown in Fig. 5 demonstrates the classification performance of the evaluated models under the SMOTE balancing strategy. Clear differences can be observed in the models' ability between classes. Among all, XGBC put the most balanced performance. For this classifier, 42 Risk regions were accurately predicted (true positives) with only 2 misclassified as safe. Whereas 21 safe regions were correctly

identified (true negatives) with only 1 misclassified as risk. The low percentage of false positives and false negatives shows to the evaluation metrics' highly balanced performance. The model exhibits high reliability in both categories and successfully separates low-risk from high-risk CBSAs. RFC also demonstrated competitive performance, correctly classifying 18 Safe and 41 Risk instances, though it made a few more mistakes. GBC performed similarly, correctly identifying 20 Safe and 40 Risk cases. ETC demonstrated comparatively higher misclassification of risk cases, especially with 9 risk instances mistakenly predicted as safe. DTC performed moderately compared to other models. The SMOTE + XGBC configuration offers the most balanced and accurate predictions for cardiovascular risk at the CBSA level. In public health applications, the low number of false negatives is especially crucial. Because incorrectly classifying high-risk areas could result in an underestimation of the cardiovascular burden. These results support the usefulness of gradient boosting with synthetic minority resampling for reliable regional health risk prediction.

4.2.8 Key observations

1. Applying SMOTE provocatively enhanced performance, increasing accuracy from 86.70% to 95.45% and rising MCC from 71.41% to 89.94%, suggesting much stronger balanced classification.
2. SMOTE+XGBC gave the best F1 -score (96.55%) and ROC-AUC (96.23) and lowest error rates.
3. Class analysis ensured observed balanced detection in both Safe and Risk classes with a 95.45% recall which led to minimum of false positives and negative.
4. XGBC also verified strong computational efficiency, making the framework suitable for scalable regional cardiovascular risk monitoring.

4.3 Interpreting models with explainable AI

In this subdivision, our study represents a comprehensive analysis of XAI for our proposed hybrid ML model SM+XGBC. XAI model framework increases transparency by finding the most significant environmental determinants and explaining their contributions at both global and local levels. By integrating Permutation Importance Analysis (PIA), SHAP (Shapley Additive Explanations), and LIME (Local Interpretable Model-Agnostic Explanations), we confirm interpretability, reliability and practical applicability of the proposed framework in environmental health risk modeling.

4.3.1 Feature importance using permutation importance method (PIA)

In Fig. 6, the permutation importance analysis (PIA) presented feature significance based on reduction in model performance when individual interpreters are randomly shuffled. The results for the top 17 positive-impact attributes reveal that atmospheric and hydrological variables control the predictive structure. Features like bare-coverfraction, lake_total_layer_temperature_1yr, water-seasonal-coverfraction exihibit the highest importance scores. These variables show the largest decline in model performance when permuted, assuring their leading contribution to cardiovascular risk prediction. The importance of bare-coverfraction proposes that land surface exposure and reduced vegetation cover play a critical role in distinguishing high-risk classes. Similarly, lake_total_layer_temperature_1yr imitates thermal dynamics in surface water bodies, which

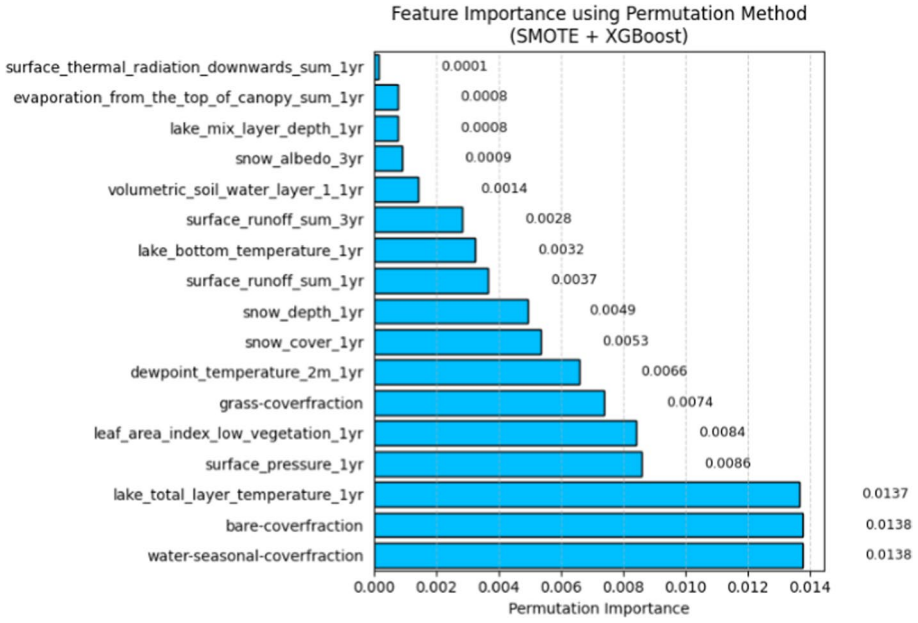


Fig. 6 Feature importance using pia

may affect regional microclimates and environmental stress conditions. The importance of water-seasonal-coverfraction further indicates that hydrological variability and seasonal water distribution significantly affect cardiovascular risk prediction.

4.3.1.1 SHAP (Shapley additive explanations)—overall impact In Fig. 7, The global SHAP summary plot expresses the magnitude and direction of feature contributions for the entire dataset. The distribution of SHAP values indicates that higher values of dewpoint temperature, lake temperature, and seasonal water coverage are positively linked to the predicted risk of CVD. At the same time lower values of these environmental factors are generally linked to predictions of the Safe class. The distinct separation of high-value (red) and low-value (blue) gradients further supports the presence of consistent directional effects. It is particularly interesting to note that multi-year temperature patterns (temperature_2m_3yr and dewpoint_temperature_2m_3yr) have positive contributions to Risk predictions.

4.3.2 Local explanation using LIME

In Fig. 8, the LIME visualization offers a local explanation for a particular CBSA that has been predicted to be at high cardiovascular risk. In this case, the features portrayed in green on the right-hand side of the visualization contribute positively to the Risk decision. Features such as layer_temperature_1yr, reservoir_content_3yr, runoff_sum_1yr, soil_sum_1yr, and precipitation_sum_1yr all increase the predicted probability of cardiovascular risk for this particular CBSA. These features all serve as indicators of the environment and serve to strengthen the model's prediction that this particular region is at high cardiovascular risk. On the other hand, the features that are depicted in red on the left-hand side of the visualization serve to negatively impact the Risk decision. Features such as snow_density_1yr, flux_sum_1yr, radiation_sum_1yr, and certain temperature-layer variables all serve to lower the predicted probability of cardiovascular risk. These features all serve as a kind of protective or moderating factor in this particular

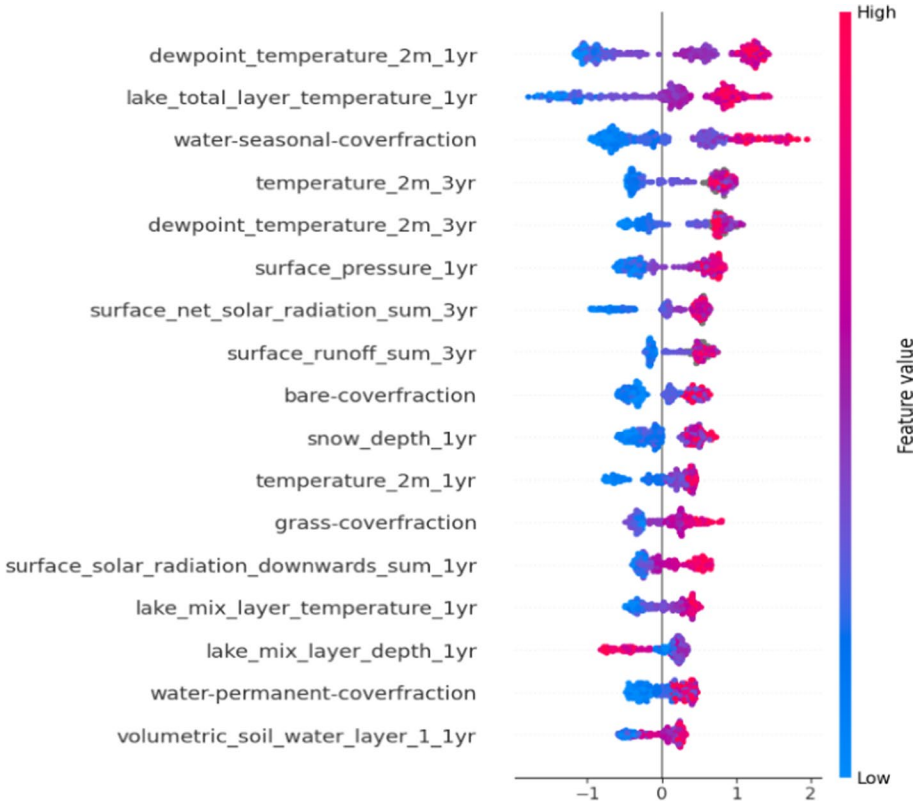


Fig. 7 SHAP (Overall impact)

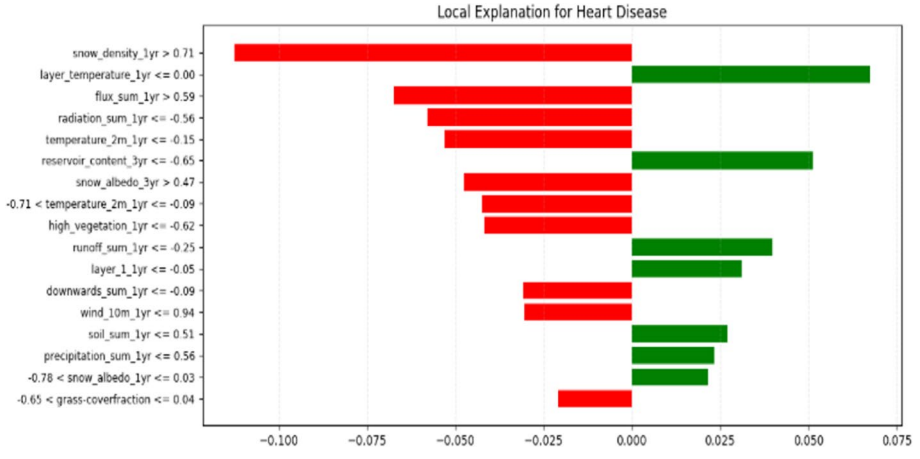


Fig. 8 Lime explanation for predicting CVD risk

case. The LIME result shows that the key positive features are constant with the SHAP and PIA, predominantly the hydrological and temperature-related variables.

4.3.3 Class-specific SHAP analysis (Safe vs Risk)

In Fig. 9A and 9B, The SHAP analysis enables a better understanding of the role of environmental factors in the prediction of Safe and Risk classes respectively. For the Safe class (Fig. 9A), the model predicts a region as having low cardiovascular risk when the environmental intensity is relatively moderate or low. Precisely, lower values of dewpoint

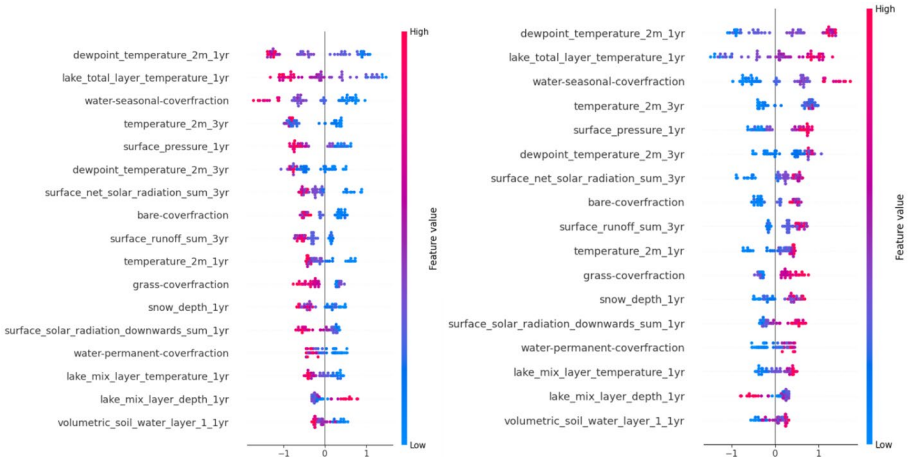


Fig. 9 Analysis of SHAP for safe and risk class of our proposed model. **a** SHAP for safe **b** shap for risk

temperature, lower lake surface temperature, lower seasonal water coverage, and lower cumulative solar radiation are positively contributing to Safe predictions. This implies that CBSAs with less extreme thermal and moisture conditions are less likely to be predicted as high-risk regions. In other words, when temperature, humidity, and environmental exposure levels are lower, the model is more likely to predict a safer cardiovascular risk profile. On the contrary, for the Risk class as stated in Fig. 9B, higher values of the same environmental factors are positively contributing to the model's prediction of high cardiovascular risk. Higher dewpoint temperature, higher lake total layer temperature, higher seasonal water coverage and higher surface pressure are strongly linked to risk predictions. This implies that CBSAs with higher long-term thermal stress and higher hydrological conditions are more likely to be predicted as high-risk regions. The same pattern for both safe and risk explanations confirms that the model is logically consistent. Higher environmental intensity is connected with higher cardiovascular risk. Overall, the SHAP analysis suggests that temperature and moisture-related environmental factors play a crucial role in distinguishing between safe and risk CBSAs. However, these feature contributions should be interpreted as predictive associations learned by the model rather than indication of direct fundamental relationships with cardiovascular disease risk.

4.3.4 Global insights from SHAP, LIME, and PIA

The XAI analysis combination has shown several global patterns:

1. Thermal and humidity-related features (dewpoint temperature, air temperature, lake temperature) have the strongest predictive power.
2. Hydrological features (seasonal water cover, runoff, soil moisture) play an important role in determining cardiovascular risk predictions.
3. Combined multi-year features have stronger influence than short-term features, suggesting the presence of cumulative effects of environmental exposure.
4. Directional consistency is evident in the model, given that the risk of CVD tends to rise with the increased intensity of the environment.

The analysis of PIA, SHAP, and LIME all prove the fact that the SMOTE + XGBC model is formed on the interpreters which are environmentally relevant and not mere correlations.

4.3.5 Overall XAI insights for the CBSA study

The explainability analysis shows that the included features related to climatic conditions, surface water dynamics and land cover substantially explain the regional cardiovascular risk patterns predicted by the model. Across the global explanation methods, bare-coverfraction, lake_total_layer_temperature_1yr and water-seasonal-coverfraction were all found to have a significant impact on the explanation. These features may relate to more general changes in the environment, including availability of vegetation, surface heat build-up, amount of urbanization, and hydrological variation, which have been also linked to variations in cardiovascular outcomes in previous environmental health studies. The SHAP analysis also suggests that some of the climate variables: dew point temperature, near surface air temperature and surface pressure, play a significant role in the model decision making process. Predictions were generally biased with higher values of various moisture and temperature variables moving towards the Risk class, and lower values towards the Safe class. The present result is in line with the overall environmental epidemiology body of knowledge on which there is a greater risk of cardiovascular vulnerability with long-term exposure to unfavorable climatic conditions. The local LIME explanations give extra details on each CBSA level prediction. In the demonstrated example variables associated with water storage, runoff, soil moisture, and precipitation had positive impact on some Risk predictions. On the other hand, opposite trends occurred at snow-related variables and vegetation indicators. Such local explanations indicate that the model is not dependent of a single environmental factor but rather reflects a combination of multiple environmental factors. Importantly, these explainability approaches look for statistical associations that the model has learned, but not at the causal mechanisms. However, the agreement among the three methods, Permutation Importance Analysis, SHAP, and LIME, gives added confidence that the proposed HXAI-ML framework is basing its prediction on environmental information that makes sense.

5 Discussion

We developed a HXAI-ML framework to facilitate predicting the possibility of heart disease given the information on the environment and health condition at the CBSA level. We assessed several machine learning models and applied data balancing techniques to lessen class imbalance and reduce prediction bias. In addition, explainable AI methods including PIA, SHAP, and LIME display the factors that influence most the cardiovascular disease prediction among the environment. The next section discusses the results of the various models and what we have learnt during the experiments.

The experimental analysis established that data balancing improved the working of the model significantly. The imbalance between the Safe and Risk classes in the data was fixed using SMOTE by creating synthetic minority examples. Combined with Extreme Gradient Boosting Classifier (XGBC), the model performed best among all the ones we tried using SMOTE. The accuracy of the SMOTE + XGBC was at 95.45, precision at 97.67 and recall was 95.45 with an F1-score of 96.55. These data were larger compared

to the results of the Decision Tree, Random Forest, Extra Trees, and Gradient Boosting. These findings reveal that synthetic minority examples when used in combination with a strong boosting model aid in learning the complicated links within the environmental and heart disease data of the model.

Models trained without data balancing performed less effectively. For example, the Decision Tree model showed the weakest performance, as it is highly sensitive to class imbalance and tentative to the majority class. The other ensemble models, such as Random Forest and Gradient Boosting were also superior but still not as good as the SMOTE + XGBC. It demonstrates the significance of resampling in the analysis of non-balanced healthcare data. This was improved by the proposed HXAI-ML framework due to the good means of resampling, high-quality ensemble learners, and explanations of AI. The SMOTE + XGBC model identified nonlinear interaction of environmental variables and maintained the classification as those of the Safe and Risk group. The integration of explanation tools such as SHAP, LIME, and PIA improved the interpretability of the predictions by highlighting the environmental variables contributing most to the model's output. This is notably essential in the health sector where healthcare providers and policymakers should understand the reason behind model's decision making. Overall, the experiments prove the hypothesis that HXAI-ML is capable of risk prediction in heart disease, maintaining a balance in the classes, and interpreting its predictions sufficiently. This renders it a reliable instrument in risk prediction of heart disease on the basis of the environmental statistics. These findings recommend that the addition of effective machine learning models with explainable AI can increase both the quality of predictive health systems and users' trust in their predictions.

5.1 Validations of research questions

RQ1 We experimented with various methods of data balancing (RO, SMOTE, combinations of each alongside TL or IHT) and evaluated them using each of the ensemble learning models. The maximum performance was with SMOTE and XGBC. Such a combination neutralized the issue with class imbalance in the cardiovascular risk CBSA-level dataset and improved the functioning of the model on new data. Its accuracy, precision, recall and F1-score were 95.45%, 97.67%, 96.55% respectively.

RQ2 Incorporation of Explainable AI (XAI) presented explicit explanations of the factors influencing the refraining of cardiovascular risk. The most important environmental variables were identified by Permutation Importance Analysis (PIA). The SHAP analysis revealed the contributions of the temperature, humidity and water-related factors to the risk predictions. LIME provided localized explanations of individual predictions, and it was useful in perceiving the way a given environment conditioned a region to be Safe or Risky. These explainability tools helped to make the model be more open and applicable to practice.

RQ3 The proposed HXAI-ML framework implements SMOTE and XGBC, is much more effective than regular machine-learning configurations that do not balance the data. It has a good balance between the Risk and Safe classes. The model achieved an accuracy of 95.45%, precision of 97.67% as well as F1-score of 96.55% and the agreement metrics

such as Kappa and MCC are also high indicating that it is reliable and balanced. These findings indicate that the model can be effectively used to predict cardiovascular risk at a region using environmental data.

Overall, the findings consolidate the research hypothesis that the proposed HXAI-ML framework incorporated advanced data balancing techniques, ensemble learning, and explainable AI. The framework effectively addresses class imbalance, making the model more interpretable, and makes prediction better. It also offers a promising approach for predicting cardiovascular risk using environmental data and may help inform future population-level monitoring and decision-making systems.

5.2 Implications to clinical practice

The study suggests monitoring environmentally associated cardiovascular risk is important for public health decision-making. The SMOTE + XGBC model outdone the other models, achieving high accuracy and balanced performance across the Safe and Risk classes. It is by far reliable in the discovery of high-risk locations in the off-balanced cases of data. The awareness of the risk location allows the health officials to prevent and take health measures in advance. The SMOTE has been applied in order to balance the data of Safe and Risk during training and the issue of uneven health data has been resolved. More fair decisions are made by the model with better balance. As well as explainable AI methods such as SHAP, LIME and Permutation Importance Analysis we have added. The model is easier to comprehend because these tools indicate the riskiest environmental factors. All in all, the proposed HXAI-ML model is a reliable and scalable method of cardiovascular risk estimation in areas. It provides enhanced data balancing, powerful ensemble models, and explainable AI to provide superior predictions that are easy to understand. The framework suggested an evidence-based approach that may support health officials in developing targeted policies and timely interventions. It has the potential to be modified for other complex environmental health challenges in the future.

5.3 Research novelty of the hybrid XAI–ML framework to predict CVD risk

The proposed HXAI-ML framework integrates information balancing techniques, ensemble machine learning models, and explainable AI. It combines environmental and population data to forecast the risk of heart disease. It is used to solve issues such as the evenness of case count, complexity in patterns with environmental data, and black-boxed nature of normal ML models. The contributions compared with the previous studies are as follows.

5.4 Linear strategic data balancing

The framework considers a number of balancing strategies for imbalanced datasets. These include Random Oversampling (RO), SMOTE, and a combination with TL and IHT. Among them, SMOTE + XGBC provides the best prediction. SMOTE generates synthetic minority scores to redress the data. This assists the model to learn reliable patterns in the environment and heart risk. It also maintains balanced detection between the safe and risk groups.

5.5 Improved explainable AI methods

The proposed HXAI-ML employs three explainability tools in contrast to typical ML studies. These methods include PIA, SHAP, and LIME. PIA demonstrates the most important environmental features. SHAP displays the general impact of every feature on risk prediction. LIME illustrates the influence of specific conditions on the classification of safe and risky by explaining each of the predictions singularly. The combination renders the model transparency and increase user confidence.

5.6 Innovation in model optimization and choice

The algorithm frameworks evaluate ensemble algorithms, including Decision Tree, Random Forest, Extra Trees, Gradient Boosting, and XGBC. Experimental analysis show XGBC with SMOTE was strongest with an accuracy of 95.45% and good precision, recall, and F1. XGBC is able to study nonlinear trends and the model remains loose. This renders the predictions excellent and rapid to large data. The proposed HXAI-ML framework will fulfill some major requirements in forecasting the risk of heart diseases based on environmental and health data.

5.7 Class imbalance

This study investigates with numerous choices of class balancing (RO, SMOTE, RO + TL, SM + TL, IHT, RO + IHT and SM + IHT). It attempts to solve the issue of not having the number of examples of the risky cases. Using SMOTE with XGBC produced the best results, artificial samples of minorities, and assists the model in learning reasonable boundaries.

5.8 Interpretability

Many AI frameworks provide explainable techniques, including SHAP, LIME, and Permutation Importance Analysis. Thus, users can understand why the model makes every prediction as well as what drives those choices. These methods provide the clear explanations of the overall and individual outcomes, which makes the model even more reliable.

5.9 Scalability and adaptability

The framework was created based on the ability to work with environmental and health data combined and can be scalable to additional regional data sets. Its overall consistency in various models and balancing approaches demonstrates that it may be able to manage large-scale environmental health risk studies.

5.10 Comprehensive experimental validation

A dataset of 329 observations and 80 attributes was tested using HXAI-ML framework based on CBSA-level data set. It experimented with a variety of classifiers and balancing strategies and compared performance to accuracy, precision, recall, F1 estimate, ROC-AUC, Kappa, MCC estimate, MAE, MSE and RMSE. The findings revealed that the best was SMOTE + XGBC that got an accuracy of 95.45% with good precision, recall and F1-score. It demonstrated that the technique is able to deal with the class imbalance besides being capable of being used to find out the complicated relationships within the data.

5.11 Practical applicability

Unlike several other models, which are mostly emphasize accuracy, the proposed framework puts an importance on how it functionality and practical application. Through XAI methods, the model requests important environmental variable such as temperature, humidity and water-related variables, which are significant and contribute to the risk of heart disease. This assists the health officials and researchers to understand better risk prediction and to help in evidence-based monitoring and action plans.

5.12 Contribution to the field

The HXAI-ML framework contributes to the environmental health risk prediction with a number of additional strengths:

- Machine learning models are used to make predictions about heart disease, resulting in a scenario whereby better predictions can be understood. The framework provides a group of machine learning models and how they choose to make decisions, meaning that the predictions can be understood and are accurate.
- The research experimented with numerous options on how to balance data and identified that SMOTE + XGBC approach in balancing data seduces the best performance over environmental heart-disease data and provides the model with more likelihood of learning.
- PIA, SHAP and LIME can be used to explain which factors in the environment have the greatest effect on a given prediction and the model itself.
- It is simple to grasp and it can be applied to a high scale as researchers and health policy makers examine how environmental conditions contribute to the increasing heart disease awareness in various regions.

Concisely, the HXAI-ML framework demonstrates the application of advanced machine-learning models with simplified expositions of how to enhance the ability of predicting heart-diseases risks using environmental data.

5.13 Future work

The framework already indicates good forecasts and can be trailed by further studies. Future work will: Trying deep learning to detect more intricate trends in large data sets of the environment and health. The data types to be incorporated during the process can be the environmental data, health records, and even opinions about the same, and similar options. Repeating the exercise with newer models as transformer-based designs to gain a better insight into how environmental data is changing over time. Such upgrades would help predict heart-disease risks even more effectively and useful in health programs in monitoring and protecting community health.

6 Conclusion and limitations

We have only used basic ensemble machine learning techniques without adopting deep learning techniques that could reflect more complex nonlinear interactions in a large-scale environmental and health dataset. Second, the framework was constructed using a pre-selected environmental variable available in the CBSA-level dataset. It did not involve deeper levels of feature fusion or multimodal integration as they may provide superior modelling results. Finally, architectures such as transformer-based deep

learning or spatial deep learning were not studied. These systems could more effectively encompass environmental exposure trends over time and space. The proposed framework also shows stable performance under 5-fold cross-validation. While the proposed framework affect showed good prediction effectiveness, it has about 329 CBSA level observations in the final dataset. Observations are regional, environmental-health data aggregated from individual patient records, but may lack generalizability due to the relatively small number of observations. Extending the proposed framework to datasets combining multiple regions and implementing other, external, validation strategies should be performed in future studies. Moreover, the proposed model will be used to assess cardiovascular risk at the area level, instead of predicting it at the individual level. The understood predictive relationships and associations might also differ between areas of different climates and different health services and populations. Moreover, the use of the framework in reality requires that environmental information available from satellites be updated regularly, and that health records across the region also be readily available in a standardized format. We should expect some variation in the extent of applicability and scalability of the proposed approach, depending on differences in jurisdictions for availability and reporting of data.

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Author contribution
Ismail Hossain: Conceptualization, Validation, Software, Writing original Draft, Writing—review & editing, Visualization, Investigation. Kazi Md. Tanvir Anzum: Methodology, Supervision, Project administration, Correspondence, Formal analysis. Jarin Tasnim Mumu: Validation, Resources, Investigation.

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Data availability
The dataset used in this study is publicly available on as SatHealth: A Multimodal Public Health Dataset with Satellite-based Environmental Factors As: https://arxiv.org/abs/2506.13842?utm_source.

Declarations

Ethics approval and consent to participate
Not applicable. This study did not involve any experiments on human participants or animals performed by any of the authors. The analysis was based entirely on publicly available, aggregated, area-level (CBSA) data that do not contain individual patient records or personally identifiable information. Therefore, ethical approval was not required.

Consent to participate
Not applicable. This study did not involve human participants, individual-level data, or direct interaction with any individuals, as it relied solely on publicly available aggregated environmental and health datasets. Therefore, consent to participate was not required.

Consent for publication
Not applicable. The manuscript does not contain any individual person's data in any form, including images, videos, or identifiable personal details. Therefore, consent to publish was not required.

Competinhg interest
The authors declare no competing interest.

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